



Music Piece Classification

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STAT 4830

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Here's a Question:

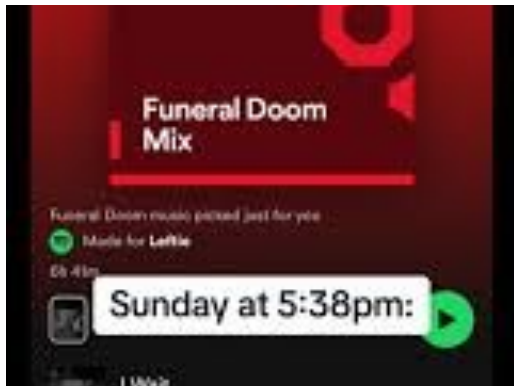
- If I played you five seconds of a Bach chorale – no context, no title – could you tell me which one it was? Out of 430?



But Why?

Real Applications:

Recommendation



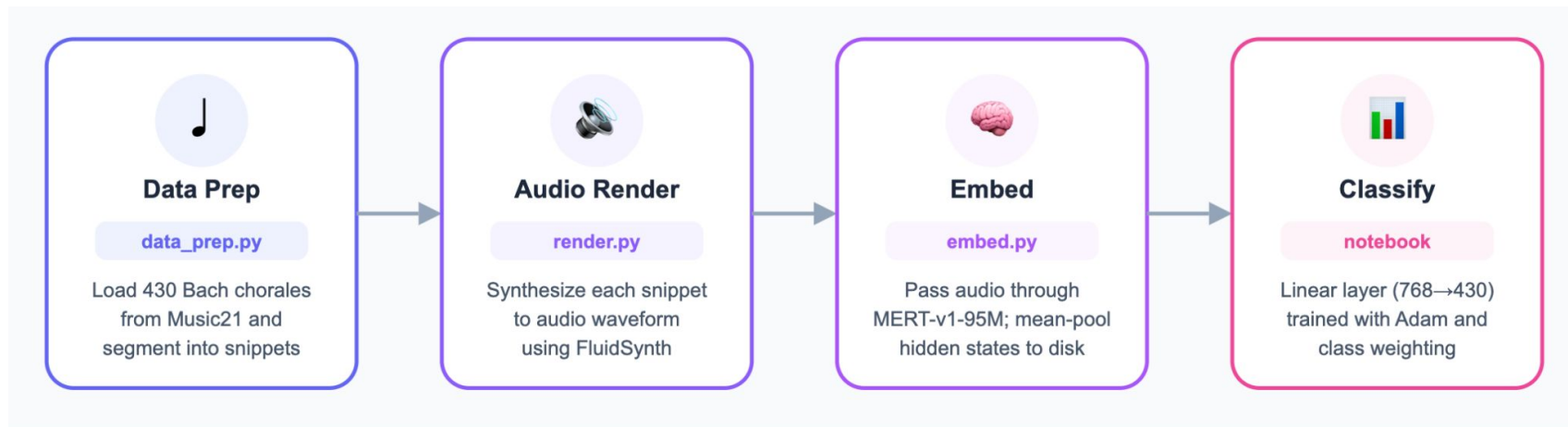
Plagiarism (Sample) Detection



Archival Search



Original Pipeline



Data Perturbation: Pipeline Additions

Can enter pipeline with music21/MIDI/WAV

1. Load MIDI: direct MIDI or music21
2. Render WAV: from the MIDI
3. Creates snippets with overlap - place all versions of a snippet in test/train
4. Perturb WAVs: using librosa
 - a. **time:** [0.8, 0.9, 1.0, 1.1, 1.2] speed up/down of tempo
 - b. **pitch:** [-2, -1, 0, 1, 2] semitones up/down
5. Embed snippets → 52775 total

Baselines of Both Datasets

Top-1 Accuracies:

	Clean	Perturbed
Random Guessing	0.23%	0.23%
Linear (logreg on frozen MERT)	34.4%	31.8%
MLP on frozen MERT	38.8%	44.2%

MLP Per-Class Analysis (Perturbed Data)

Total: 49 classes with 100% accuracy, 114 with 0%

Summary of errors:

- Same chorale, different harmonization
- Same key, similar/different rhythms ← majority
- Different key, similar rhythms

~8.8% chance of same key between 2 pieces

vs.

45.38% fraction of misclassifications were where key was same, see example →

BWV 437



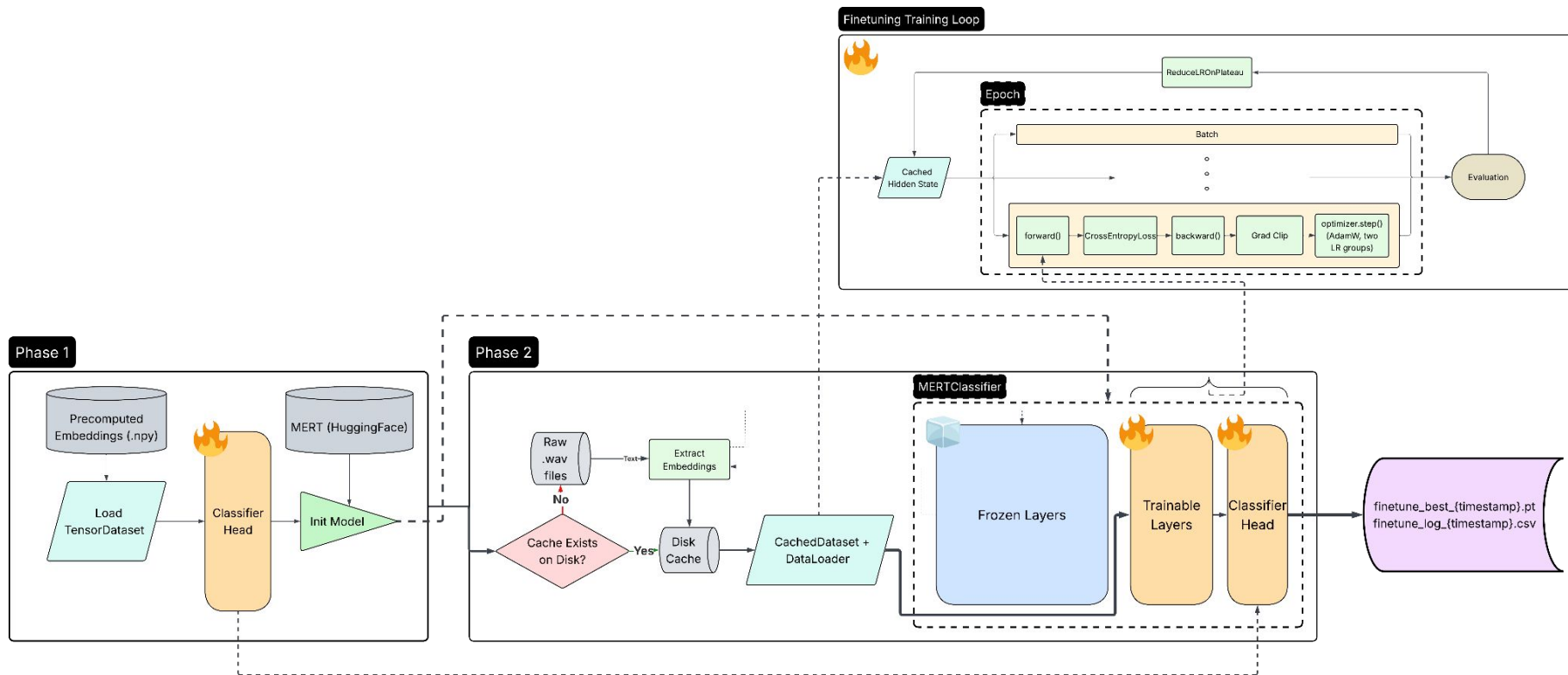
BWV 276



Finetuning - Goal

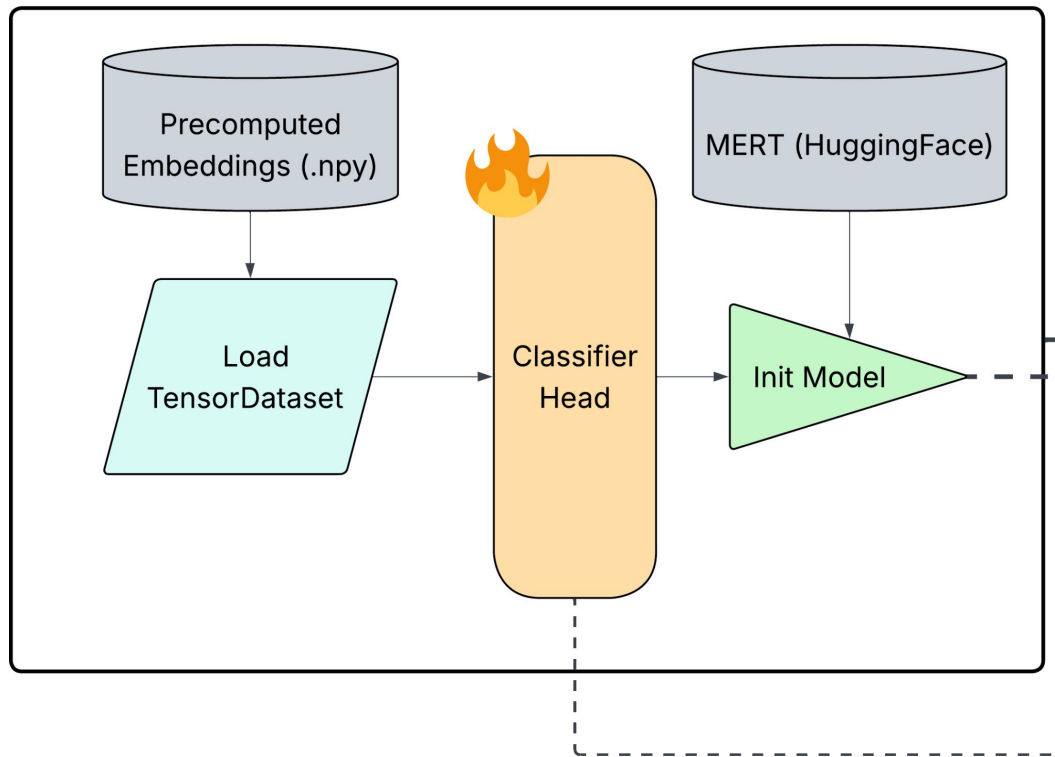
Goal: To finetune some transformer layers of MERT using our data to adapt how it creates embeddings to support our classification purposes. By doing this training along with a classification head with weights from previous well-performing MLP, it provides a better starting point.

Finetuning Pipeline

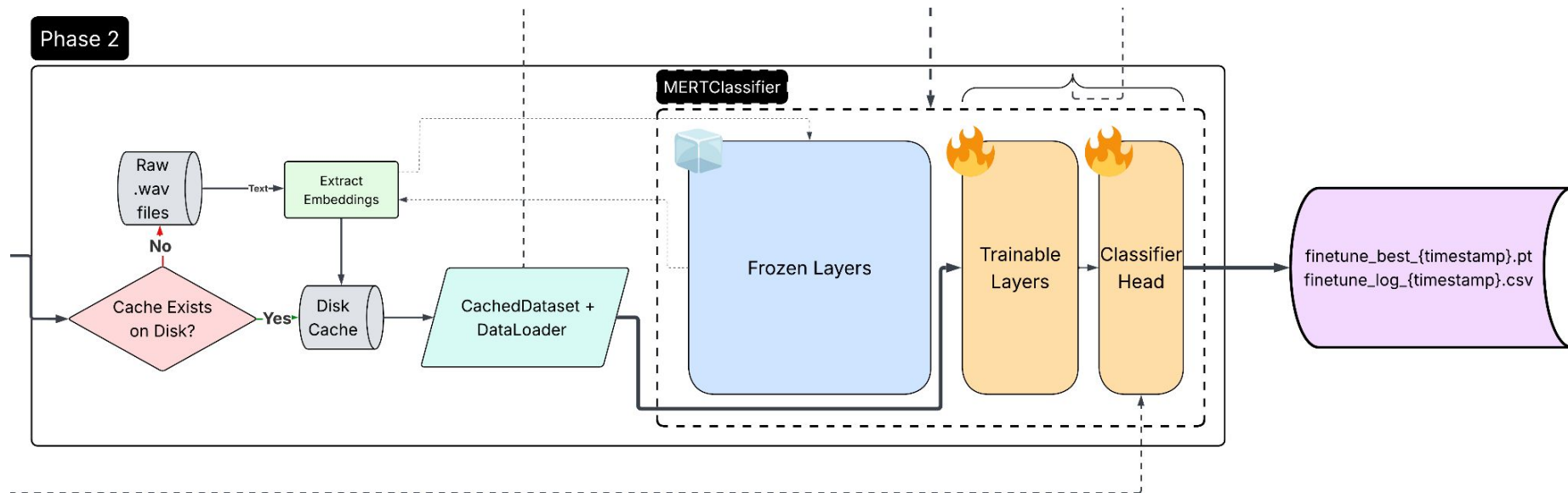


Phase I

Phase 1



Phase 2



Finetuning Results - Clean

Architecture

4 layers unfrozen

[512] + 0.3 dropout

Accuracies

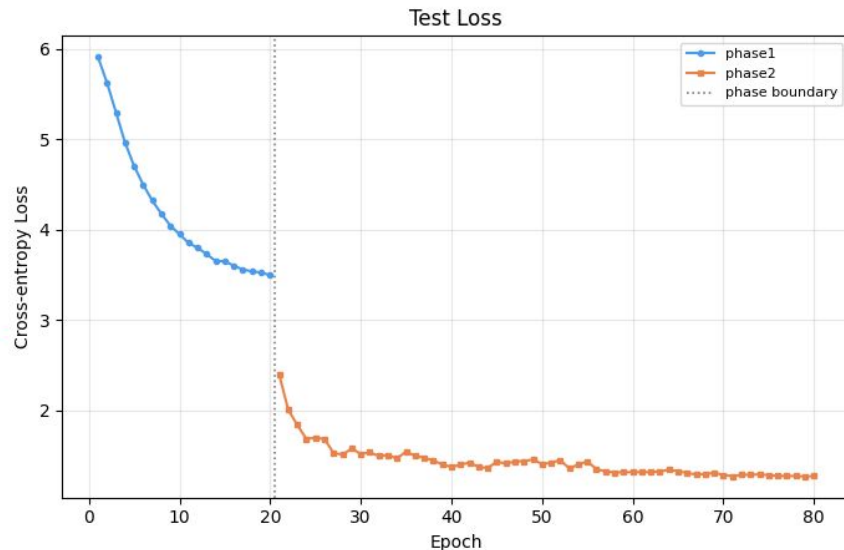
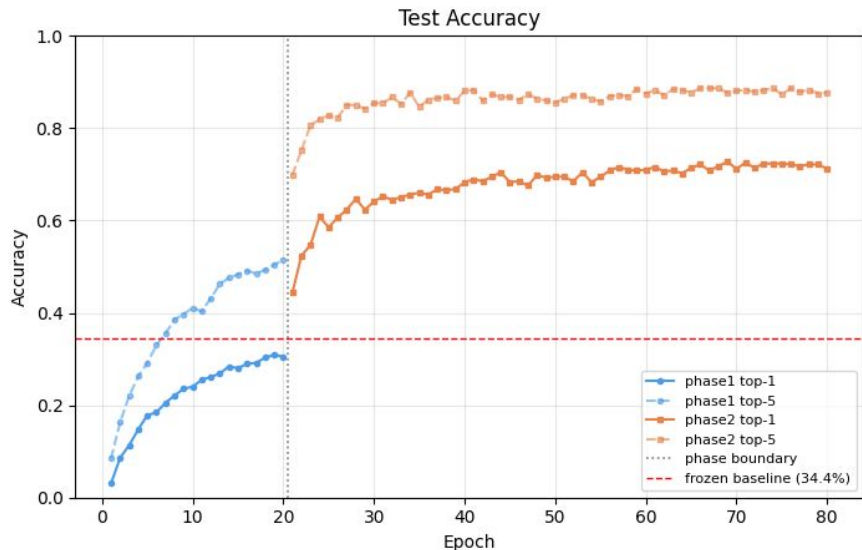
final top-1: 71.2%

final top-5: 87.7%

best top-1: 72.7%

best top-5: 88.7%

Fine-tuning Training Curves



Finetuning Results - Perturbed

Architecture

4 layers unfrozen

[2048, 512] + 0.5 dropout

Accuracies

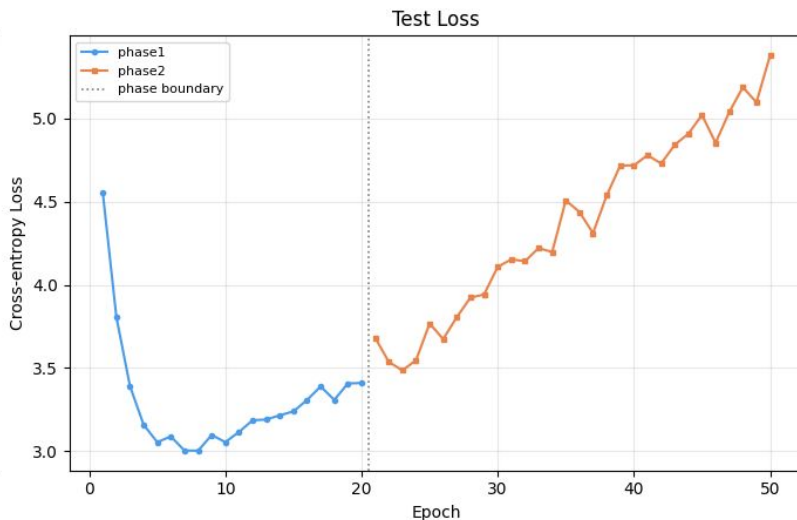
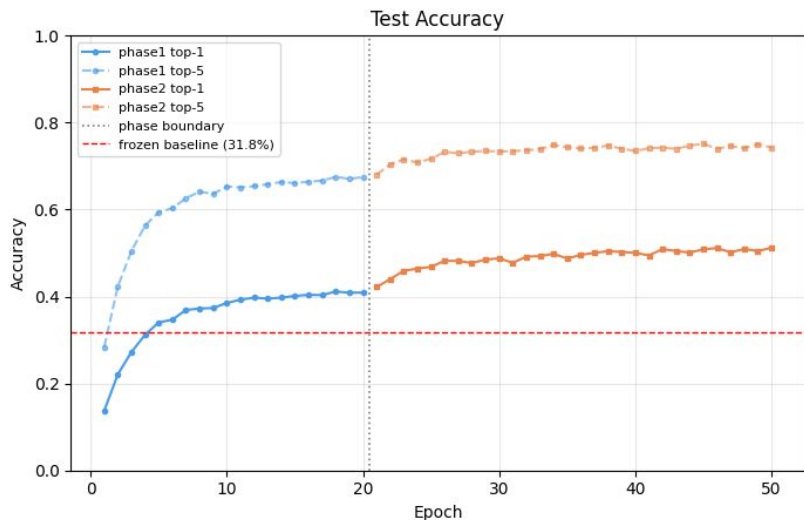
final top-1: 51.2%

final top-5: 74.3%

best top-1: 51.2%

best top-5: 75.1%

Fine-tuning Training Curves



Dataset Cross Experiment

Results

	Test: Clean	Test: Perturbed
Train: Clean	72.7%	12.9%
Train: Perturbed	53.8%	51.2%

Takeaways

- Clean-trained models fail with tempo/pitch variation
- Perturbed-trained models are robust, at a cost to clean-only performance

Finetuning - Time Aware Models

Problem: Phase 1 precomputed embeddings are mean-pooled

- Regenerating embeddings is expensive in both time and disk space
- (T, 768) dimensional embedding $\rightarrow$ single 768 vector
- T for an 5 second snippet is 374 frames (24 kHz)
- If our current embedding file is 150 MB, our new embedding file would be > 50 GB
- We observe this size differential when caching embeddings in Phase 2 of our pipeline
- We're currently working on this, but we chose to devote to other experiments first.

Temporary Solution: Skip Phase 1?

- Tested basic BiLSTM and Transformer heads as well as Attn Pooling on MLP final layer
- Transformers perform abysmally with BiLSTM being approaching the baseline and Attn Pooling approaching MLP performance

Conclusion

- 430-way Bach chorale classification using MERT + a trained head, top-4 MERT layers finetuned.
- 72.7% top-1 on clean audio, 51.2% on audio with 25 tempo+pitch variants.
- The accuracy drop from clean to perturbed came from pushing different snippets' perturbations closer together

MLP results in summary

Unperturbed (no overlap)

- 2487 train, 521 test snippets
- **0.344 top-1 accuracy, 0.564 top-5 accuracy** (1 hidden layer MLP [512])

Revised perturbed v1 (41850 train, 10925 test snippets) - fixed data leakage

- tempos [0.8, 0.9, 1.0, 1.1, 1.2], pitches [-2, -1, 0, 1, 2]
- 8s snippets, 2s overlap (i.e '6s hop')
- **0.4422 top-1 accuracy, 0.6767 top-5 accuracy**

Revised perturbed v2 (75330 train, 19665 test snippets) - v2 of above

- tempos [0.8, 0.9, 1.0, 1.1, 1.2], pitches [-4, -3, -2, -1, 0, 1, 2, 3, 4]
- 8s snippets, 2s overlap (i.e '6s hop')
- **0.4287 top-1 accuracy, 0.6608 top-5 accuracy**

MLP Search (Revised Perturbed v1)

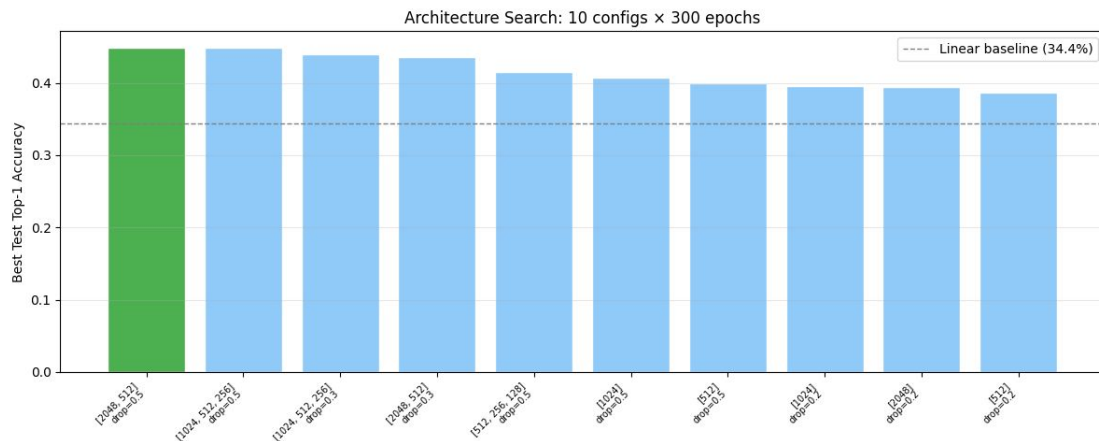
	hidden_dims	dropout_p	n_params	best_test_top1	best_test_top5	final_test_top1	final_lr
1	[2048, 512]	0.5	2,836,885	0.4491	0.6781	0.4457	1.25e-04
2	[1024, 512, 256]	0.5	1,551,253	0.4483	0.6930	0.4445	2.50e-04
3	[1024, 512, 256]	0.3	1,551,253	0.4390	0.6778	0.4363	1.25e-04
4	[2048, 512]	0.3	2,836,885	0.4362	0.6662	0.4340	1.25e-04
5	[512, 256, 128]	0.5	611,989	0.4149	0.6891	0.4056	1.00e-03
6	[1024]	0.5	1,204,629	0.4073	0.6294	0.4046	1.25e-04
7	[512]	0.5	602,517	0.3993	0.6155	0.3959	1.25e-04
8	[1024]	0.2	1,204,629	0.3952	0.6085	0.3928	1.25e-04
9	[2048]	0.2	2,408,853	0.3941	0.6032	0.3829	5.00e-05
10	[512]	0.2	602,517	0.3866	0.6094	0.3866	3.13e-05

Best config: [2048, 512],
0.5 dropout

After retrain w/ 1000
epochs:

Final test top-1: 0.4422

Final test top-5: 0.6767



Finetuning

- **Phase 1:** Train MLP classifier on precomputed embeddings
 - $768 \rightarrow \text{BatchNorm} \rightarrow 512 \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow 430$
- **Phase 2:** Unfreeze top n layers of MERT, using weights from Phase 1 as starting point, train end-to-end with audio WAV files (i.e. train those MERT layers in combination with following MLP)
 - Have tried top 2 and top 4 layers
 - Train for 30 epochs - save .pt and log

Finetuning Results - Perturbed Dataset

2 layers unfrozen
[5|2] + 0.3 dropout

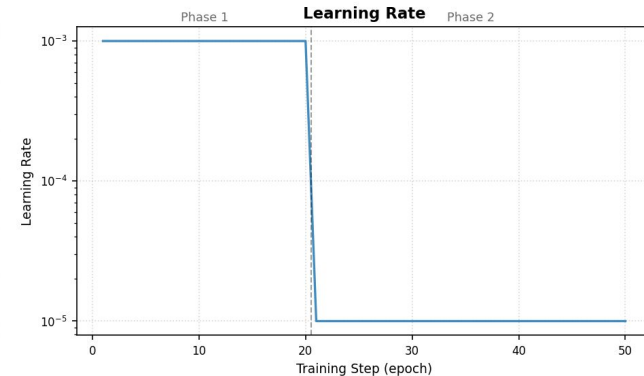
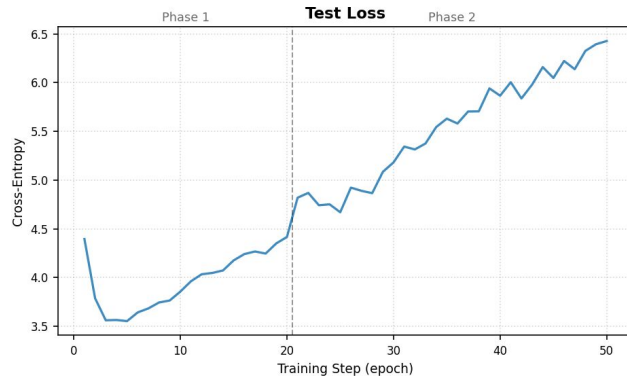
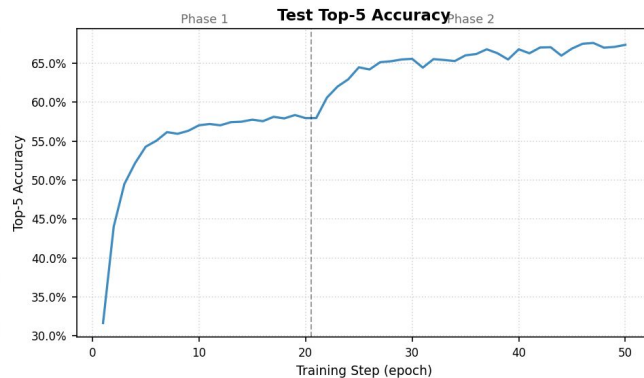
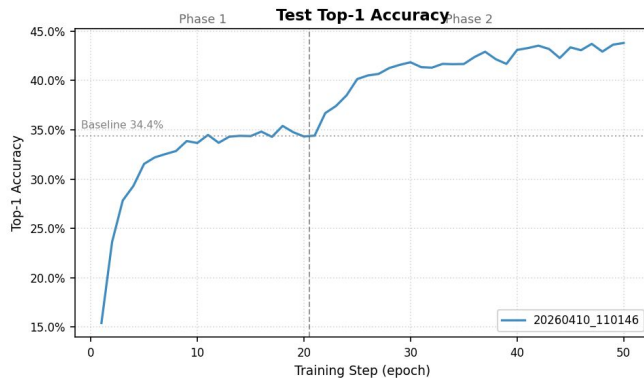
Accuracies

final top-1: 43.8%

best top-1: 43.8%

final top-5: 67.4%

best top-5: 67.6%



Finetuning Results - Perturbed (1 layer architecture)

4 layers unfrozen

[512] + 0.3 dropout

Accuracies

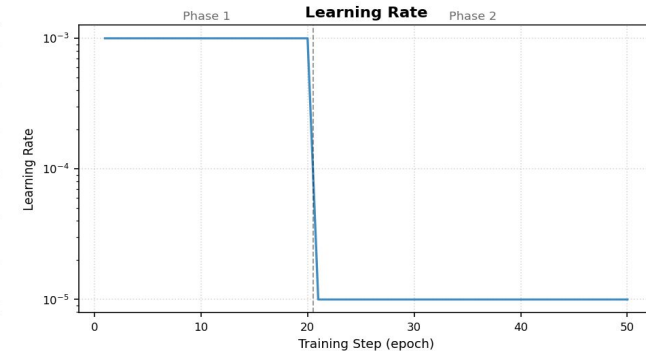
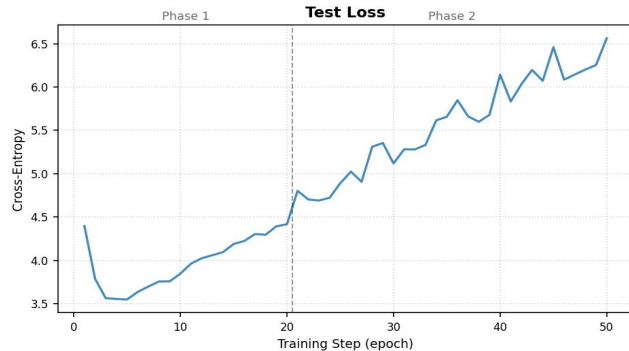
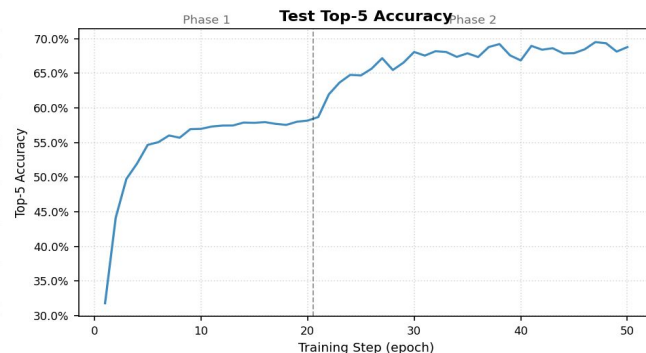
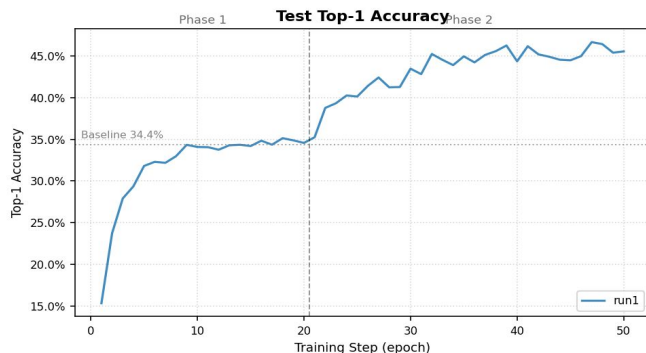
final top-1: 45.6%

best top-1: 46.7%

final top-5: 68.8%

best top-5: 69.5%

Comparable to just
MLP on perturbed



Fine-Tuning vs. Perturbation Robustness

Setup

Two fine-tuned MERT models (2-layer unfreeze, same architecture) trained on different data:

- Model A: trained on 2,487 clean snippets
- Model B: trained on 41,850 perturbed snippets (25 tempo/pitch variants)

Experiment I — Does perturbation training help robustness?

Evaluate both models on the perturbed test set (10,925 snippets, all 25 variants).

- Model A result: 12.9% top-1 / 20.1% top-5
- Model B result: 46.7% top-1 / 69.5% top-5

Fine-Tuning vs. Perturbation Robustness (cont.)

Experiment 2 — Does perturbation training hurt clean accuracy? Evaluate both models on the 437 clean (t1.0p0) test snippets.

- Model A result: 68.9% top-1 / 86.0% top-5
- Model B result: 53.8% top-1 / 74.8% top-5

Interpretation

If Model A degrades significantly on perturbed audio → fine-tuning on clean data learns audio-specific features that don't transfer across perturbations.

If Model B underperforms Model A even on clean audio → perturbation training disrupts the fine-tuning signal without compensating with robustness.

Accuracy Drop Explanation

